Active Inference LiveStream 006.0: "A tale of two densities: active inference is enactive..." (2020)

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all right i hope we are live hello and welcome this is active inference live stream 6.0 it is october 9th 2020. i am daniel friedman and i'm going to be doing a solo contextualizing episode i guess of active inference live stream today on the paper tale of two densities welcome to team com we are an experiment in online team communication and learning related to active inference you can find us on twitter at inferenceactive at gmail via our public key base team or via youtube which is where you may be watching us now this is a recorded and an archived live stream so please provide us with feedback so we can improve on different aspects of our work all backgrounds and perspectives are welcome here and as far as video etiquette goes mute if there's noise in the background raise your hand etc here we are in actim stream 6.0 and the goal of this stream is really to set the context for the upcoming discussion next tuesday on 6.1 that discussion is going to be on the paper a tale of two densities active inference is an active inference by ramps that at all published in 2020 online in 2019 and what this video is going to do is provide context for some of the ideas types of models philosophical threads and vocabulary of the ramstead at all paper in other words it will help contextualize why this topic is being discussed why there's a regime of attention on this topic so to speak here are the sections of this video first i'm going to frame the problem as being about what the free energy principle and active inference say about agents and the world and their relationship and then to contextualize different ways of thinking about that we'll learn about bayesian models and inactivism just brief brief takes of course you could go a lot into depth on different takes here i'm just trying to really cut in broad strokes and then in the last part of the discussion today we'll talk a little bit about the philosophy of science as far as the difference between a principle a theory and a hypothesis within the first in sphere using uh his response to an interview question so next time on uh number 6.1 we're going to discuss the paper with anyone who wants to participate so please save and submit your questions if you're interested in participating or if you're watching it after ask your questions then or if it's before get in touch with us if you want to discuss all right so here it is the real topic that we're going to be addressing here is what does the free energy principle and active inference say about the relationship between the world and agents and how we think about this formally in terms of how we model it with computers or just cognitively as well as how we experience it it's just all important so this is a pretty general topic and so here's one example of a world and an agent but world and agent are not only used interchangeably as we'll later find out because one one agent's world is another world's agent and so on but really just because this is a pretty general framework of relational modeling so it doesn't have to be you know the the globe or the world as you exactly think of it and agent as in a robot agent we're going to be probably thinking broadly about what it means to be connected for two things and this video certainly is just an introduction to the context of some of these ideas it's not a review or meta-analysis or final word it's just as i was reading through the paper it was how i structured the information and i was how i was uh thinking it might make sense to read some of the quotes and use those as springboards to talk about the bigger questions uh as usual trying to start from the active inference perspective and then reach out to the broader questions as well as uh go the other way i guess and uh in this video specifically as it relates to free energy principle i'll just provide the punch the punchline first the punch line is that generative models are not structures they're embodied and enacted regimes of policy attention and behavior so if it doesn't make sense to you yet why that's the punch line or why that's one of the conclusions essentially of this paper then just keep listening and learning sorry so in this talk specifically the structure is going to be like this first we'll go through the goals of the 2020 tale of two densities paper read through the abstract then we'll talk about bayesian inference models kind of steel manning them putting them in a strong light then talk about inactive and action models pretty broadly though there's a lot of nuance in both of these categories then talk about agent world modeling in fep and active and previewing mainly by reading quotes from the paper and highlighting specific claims um highlighting how active inference and free energy principle build on this uh map and territory that are in play with bayesianism and activism and again that underlying question is what do we have to say about how the world and asians and agents and agents are related to one another all right so the paper is called a tale of two densities a literature reference there active inference is an active inference by ramstead kirchoff and fristen in the journal adaptive behavior in 2020 and the goal of the paper which i believe is the first sentence of the article is the aim of this article is to clarify how best to interpret some of the central constructs that underwrite the free energy principle fep and its corollary which means related theory active inference in theoretical neuroscience and biology namely the role that generative models and recognition densities play in this theory aiming to unify life and mind and they'll be a consistent color scheme for the generative model and the recognition densities because um we're going to be looking to integrate these two different ways of thinking about agent world dynamics in the bayesian and the inactivist worldviews and so what we're going to talk the most about here are what these two densities are so you can ask in the book what were the two cities here we're going to ask what are those two densities then we can ask what is this tail that integrates these two densities what weaves them together into one narrative which helps us understand and uh helps us learn so let's go first to the abstract we already read the title which is the first way that the authors want to represent their work now we're going to look through the abstract and then later on we'll end up defining and working through a lot of the words specifically but starting with the abstract is the first thing that you're going to read about the aim of this article is to clarify how best to interpret some of the central constructs that underwrite the free energy principle and its corollary active inference in theoretical neuroscience and biology that may even be what we already read so same meaning there and variational densities play in this theory so those are the two densities that we're going to be talking about we argue that these two constructs which are the two densities the generative models and the variational densities have systematically been misrepresented in the literature because of the conflation between the fep and active inference that's why at the end we're going to be really specific about the difference between those two on one hand and the distinct albeit closely related bayesian formulations centered on the brain variously known as predictive processing predictive coding or the prediction error minimization framework and we'll return to how the prediction or the predictive brain is uh related or different as a from a philosophy of science perspective and from a computational perspective i guess all right second part of the abstract more specifically we examine two contrasting interpretations of these active inference type models a structural uh or really variational densities the things that we're eventually going to see brought together an active inference these okay i i believe but if it's unclear i'm not sure a structural representationalist and an active approach and that's going to be our whole main discussion we argue that the structural representationalist interpretation of generative and recognition models does not do justice to the role these constructs play in active inference under the fep so the main theme that's going to get returned to is that the computational side is very tractable stable however it misses some fundamental components that are going to be really complementary to the insights that are brought from the inactive approach we provoke we propose an inactive interpretation of active inference what might be called an active inference in active inference under the fep the generative and recognition models are best cast as realizing inference and control the self-organizing belief guided selection of action policies and do not have the properties ascribed by structural representationalists so that's just saying structural representationalists are wrong about properties in what they think because of actually generative models being basically cybernetic policy selectors their claim is that that makes some of the stances held by structural representationalists untenable so let's go to the paper here's a quote the question that shall occupy us is how best to understand the function and properties of the generative and recognition models in active inference under the fep what does it mean for one theory to be under the other we'll return to that at the end in light of the active processes involved in orchestrating maintaining and updating these models in particular we examine two contrasting interpretations of these models a structural representationalist that's one of the schools of thought here and an inactive interpretation so two different interpretations so we can summarize this up as a structural representationalist versus inactive whose hot take on systems will reign supreme so first let's just try to understand what each of these groups are really bringing to the table because uh that's really hopefully a strong starting point from each of these two sides can lead to a better synthesis in active inference so in bayesian inference it's a big area this is just one way of presenting it you can definitely watch other lectures or learn from a lot of other more skilled people in this specific area about bayesian approaches so this is just one representation i was trying to make a compromise between things that were implemented but there's so many flavors and colors of what's implemented in the literature so i just wanted to start with a really brief overview of the bayesian inference this two level bayesian inference scheme so variously these two elements are known as priors and hyper priors or priors in the data but here on the left side we have data which are reflecting sensory observations like that of a sensor and then there's a generative model for example that generates sensor readings and so the the question that is asked in the transformation from the sensory observational data the lower level model on the left side to the generative model of the data which is like often the higher order parameters in the computational model the question is how do these observations influence or update the generative model of the world and so you can think about that as sort of like empirical data coming in altering your parameters that represent deeper stances about the world so you have a sensor that fluctuates more rapidly and then you're doing some sort of uh stabilization of the overall readings of the sensor and that's like your higher order parameter just to give one of many samples and then the other question that's being asked in the transformation from how the parameters of the higher order model influence the data is basically what observations are emitted here given the generative model of the world or the generative parameters of the world which includes different things like world states causes etc so here's some of the features and advantages of bayesian models so first they can extend to basically arbitrary graphical models which doesn't just mean models you can draw it means models that you can graph so it turns out that a lot of classical and uh non-classical statistical techniques can be reduced down to this type of a scheme at the highest level statistically it's discussed in carl firsten at all's textbook related to spm how these kind of bayesian inference models relate to various empirical data sets so potentially of interest for people who want to learn more about that side the next thing that's good about these models is that they can be used to model multi-scale systems so through space and time of different scales they have very tractable computation and implementation on hardware so you can take a data set and basically get useful information out of it using this type of a scheme so it passes the sort of utility check it's pretty flexible with having sparse or dense multivariate data and i'll tie that to the second point here which is that you can be empirical in the sense that you're learning and updating your position based upon data but also you're formalizing the use of your prior information you're neither just informally combining your prior information with your uh newer information naively just saying oh well they're both equivalent because i have one of each one of the past one of the present uh it provides more formalized ways of dealing with with information and depending on how much you rely on your prior information you may only need very sparse inputs from your sensor for example you might only need a thermometer in a certain location to know whether it was in one climate regime or another these models can also be fit with expectation maximization algorithms that's also discussed a lot in fristen's spm textbook and uh work just and in the work of people who have done a lot more on expectation maximization algorithm um learning then i'm going to go into detail in here but it's a good topic to look into and one that i'm still learning about too uh kind of as stated in a few points these models are pretty deeply related to statistics of various kinds as well as other mathematical areas so let's just go to the structural representationalist interpretation first structure i think that there's probably wide latitude in what the structures mean and there were several citations that characterize the structural representationalist viewpoint so i'll let them speak to what structural representations and structural representationalism is i would say structures could represent data structures it could represent things having structure in the sense that they're distinguishable from something else so structure could be pretty broad representations that's a whole topic that i understand there's also a lot of debate over and we've read some other act inflated papers on representations and uh their strengths and weaknesses i i just think it is what it is and this structural representation is holding the views that the authors have in their citations and that's where we'll sort of end the discussion on the bayesian inference models and think about these are all really good features of the model what is going to be something that's going to complement that from a pragmatic and philosophy of science perspective so the second category of models is this inactive or active models and here it's a lot more focused through a tradition of being action oriented and ecological it's more oriented about the relation between the world and asia so this comes from a lot of areas like systems and ecology so the two questions that are being asked in the inactivist perspective are first off how the world is influencing the agent is a question about how perception is influencing or updating the action tendencies of the system and then the way that the agent is influencing the world or that causal arrow is kind of asking what actions are entertained by internal models and entailed through control mechanisms of the agent so both of the questions are about the joint niche between the world and the agent which will be formalized later we'll find as a markov blanket but for now we can just think about the world and the agent being connected through this perception action loop so what are the cool parts about the or the active perspective so it's interdisciplinary which encompasses a lot of interesting work in areas like art cybernetics ecology psychology dynamical systems sociology and tech and evolution so kind of also things that the bayesian models have been applied in so it's not unique to this but some of the work in these areas is very interesting there's a qualitative and a relational and experiential aspect and anything in the computational domain it's usually a question about efficiency whereas things that are in the qualitative domain often it's about concordance with our experience or potentially even other things so uh it's an interesting and different development of theory to go down the route of more in touch with phenomenology phenomenologists experience per se rather than more implementable on modern computational hardware it naturally relates to things like enculturation developmental processes sociality one agent the world could be the body of another agent or it could be a small community or something like that it turns out that this inactivist perspective is more amenable this externalism which we discussed in a previous active stream and other kind of radical embodied cognitive theories so i i'll just leave that there that's kind of more of a connector but again there's more and fusion between these two approaches as evidenced by the discussion that we're having today but these inactive approaches have definitely a very different um literature base different citation networks that are linking these ideas so i'm just phrasing them in a way that's not representing priority or importance just sort of throwing out different features to differentiate it strongly from the bayesian perspective computational all right inactive models also interestingly and this is one of the reasons why they've recently received a lot of attention is that they're amenable to different kinds of things that maybe people would call physical computers in 2020 like ant colonies slime molds landslides quantum systems all these different things that uh do what can only be described as special types of algorithms or computation yet they don't do all problems so they're they're more like a bitcoin miner than a uh turing complete processor and computing language so they're specialized computers if they're computers personally i don't think the computational metaphor applies so well because i also think that this is a perspective that takes us beyond computationalist metaphors of the world sometimes but at the very least the inactive perspective highlighted all these ecological dynamics of intelligence that are only now being appreciated from a more quantitative and also these active models as you might guess are part of a broader tradition of action and pragmatism so it's a another interesting area that kind of intersects there so here's the simplification slash uh side by side first i'm just gonna again try to distill it another level down without uh leaving too many false positives or too many false negatives so if someone knows better about these areas totally i'm open to just correcting it and changing it i'm just as i read through the paper i just mock this up to try to understand how these theories are different and similar so a sort of quasi-bayesian structural representationalist view we can distill down to this we can say that there's the data parameters and the hyperparameters or just the parameters of one model and that of another and this can be again stacked up multi-level all these different things that we discussed earlier and the way that the data like coming in from the sensor update the hyper parameters like the beliefs about the world are through what is going to be called the recognition model by that community on the other hand the way that the predictions about the world state are used to generate the data parameters is a generative model and that is just like also considered sensory inference now it turns out that there's a lot of model and inference being thrown around like isn't everything a model and all this sort of stuff yes it does um get a little confusing and so i've tried at a few points to be really clear and specific but sometimes in quotes um they uh will use like model or who's doing inference where and we're just going to be pretty specific about what all the connections and the pieces are and then hopefully it'll be clear um what yeah what's similar and different between these two approaches so the outcome of the bayesian model like a successfully converged model that's running here on a computer it represents a statistical converge multi-level model that represents the structures of the world so let's just think about the data parameters like the visual field like the sensor data on the retina so to speak and the hyper parameters being like where objects are and so even though there's no color vision in the periphery and even though there's low resolution vision in the periphery still i'm able to have a consistent uh relationship things look pretty normal the sensory data are being generated in basic non-surprisal with the recognition modeled sensory data so that would represent a sort of bayesian computationalist perspective on vision and error correction and optimization on the other hand we can side by side with the inactive perspective so the inactive perspective again situates the agent in the world as being linked through perception and action and the outcome of the inactive perspective is an embodied and ecological action sequence or like some sort of niche interplay from an embedded agent who makes or is behavior and this is a lot of the inspiration for things like plants intelligence or something like that it's just oh well what time scale and what problems are you asking the system to solve and all these things that ultimately are the decisions and the classifications of the scientists or so it said uh are used to talk about the different intelligence or perception or action capabilities of different agents so let's try to unite this under what is said by free energy principle and active inference in this gap explanatory and causally between the world and the agents so on the top of this slide there is a figure one from the previous paper we discussed in five i believe and several other papers and it describes this minimal loop between the organism and the environment so this is the first bridge uh connector point so if you're on board with just the questions about the world and the agent then we're still talking about just exactly that and we want to build on the work and the perspective of the different insights from a few different areas first from the schools of action learning about things like inactive embodied ecological practice socio-technical and experiential uh aspects of the inactive view that we discussed a few of earlier from the more inference side again there's a lot of directions which inference is being used in but it's one of the words um in active inference and so that's why i'm using them here and on the more inference side we can think about drawing the insights from bayesian and computational approaches statistical approaches modeling uh deep learning robots so all these kind of interesting things that think of the world whether uh conceptually or just from a scientific modeling perspective from a computational perspective and then also there's uh another level that as we'll see is related to the roots of active inference in the free energy principle but there's this whole root layer of physics and specifically hear things like invariant sets and manifolds patterns multi-scale dynamics collective behavior phase transitions as well as potentially a bridge to things like unconventional computing or more advanced or more tractable world modeling so here we've gone one level deeper into that world agent mapping so we still have world on the in the teal and then agent on the right side in the darker blue except they're labeled now as internal and external states and they're linked through these two specific types of states called sensory and action states so i'm going to go through a list just features about the fep and active sort of skeleton of how the agent in the are related to and then see if that covers this diagram on the bottom left we'll see so internal and external states so that's respectively agent in the world states are linked through situation and model dependent blanket states so it's not like there's just one simple barrier there is a something that is contextualized that's a functionalized barrier in the context of the organism or in the context of the scientists it and so the states can be coarse or fine-grained and that's really a map territory situation so uh depending on how coarsely grained success in a situation can be uh basically achieved then even very corresponding of certain strategies will work for example and one thing that we'll return to again and again is that active inference is about this relationship between the sensory input and the action selection not necessarily just the convergence on the hypothetical model it's about the so um there's this dual instrumentalism and i think it was a really nice sequence of points brought up by maxwell and others in the one of the five discussions about this dual instrumentalism and so to kind of rehearse what those two instrumentalisms are so the first usage of these types of partitioning models is as a scientific framework because any system of interest you can identify you can identify its inputs and its outputs and its internal states and its external states and you can almost think well what would what would make this false how could i be outside the system and model it in a way where there was internal states that um influenced external states without or vice versa so that might be something that we don't know how to measure or it could happen but if it's a part of your model you could specify it with a different topology we'll return to the question of whether you could specify different partitioning and have it all work out but i'd suffice to say that for now we're just thinking about the skeleton of this relationship between the different states and as far as modeling from a scientific framework perspective goes you can basically partition any measurable system into this kind of a the part that was referred to as speculative in the previous paper was as to whether this is actually what organisms do and that's sort of a philosophical area that let's touch on later on uh and in ongoing discussions because i think that's sort of one of the areas that we return to uh almost every week in acting extreme so just look at this diagram think about whether you agree with the way that states are partitioned and keeping in mind the multi-scale insights from some of the previous discussions as well and could there be other ways that these are connected and then we're going to want to specifically draw together the bayesian and inactivist ways of thinking about action okay so here we go to the section where i'll read some more quotations from the paper and just highlight draw their exact argument about what the different schools of thought bring into play so first they characterized the structural representationalists as having a good take on their own models but missing action as a fundamental consequence of a real far from equilibrium agent existing in the world so the primacy of action means that reading from the second paragraph of generative and recognition models these two bi-directional arrows while providing an accurate description of these constructs as they figure in some version of bayesian cognitive so in other words being a disciplinary tool does not do justice to the generative models and recognition densities that figure in active inference under the fep so in other words the interpretation of how those two types of models agents and world and models and hyper models how those two things are related uh the computationalist perspective or the the so-called structural representationalist perspective uh it's missing some aspect uh and then contrasting with the second red underlined part here in contrast to these other bayesian i.e purely bayesian theories which are in effect theories of the structure function and dynamics of the active inference is a much broader theory of adaptive phenotypes that centers on the control of adaptive behavior and that emphasizes the tight coupling and circular causality between perception and action so that's the part that draws it more naturally to systems thinking and to cybernetics um all right so structural representationalists argue that generative models function as structural representations so no surprises there with representational content okay a great summary of this view which is a nice thing to say about the authors reads that predictive coding theory and this is being pop provided as a specific kind of sub model within this bayesian structural representationalist paradigm which we'll return to at the end as far as their formal relationship predictive coding postulates internal structures so that's where you get the structure apart who's functioning inside of a cognitive system closely resembles the functioning of cartographic maps cool it might be said that on the proposed interpretation of the theory cognitive systems navigate their actions so is it not action-oriented through the use of a sort of causal probabilistic maps of the world so there is a possible uh place for a structure so it's the maps are the the representations that are in the uh neural system these maps play the role of representations within the theory okay believe it or not i guess it so specifically this map like role is played by the generative model it is the generative models that similar to maps constitute action guiding detachable structural representations that afford representational error detection so here's what they're proposing they're saying that generated maps from some sort of map generator function are used to then do policy and agenda setting so action gets separated from second layer estimation in this way that's characterized as in that right side of the third line from the bottom so it's uh that's the perspective of the authors of this paper characterizing hopefully in a positive light the structural representation lists so those are all interesting things i think that are cool as well as all the aspects of these models that are beneficial discussed earlier so on the other hand let's look at what inactivism brings to the table in this section we unpack the implications of the pragmatist view for understanding the relations between the the generative process and the recognition model according to our pragmatist interpretation the organism embodies the recognition density and entails the generative model as a control system we then formulate a direct critique of the claim that generative models are structural representations so they're going to say because of embodied enacted whatever it is the generative model is a control system and then because of the control system and not a map of the world like where are the structures not where the processes of the world there's no google maps for processes yet but they're going to formulate a critique in the end that generative models are not structural representations because they are control systems and control systems are not structural representations they're something else process realizations or something so here's what inactivism ends up adding it ends up providing a pretty natural pragmatic and action-oriented perspective on active inference modeling it brings all of these 4e 5e cognitive perspectives in this is really an interesting area of research maybe there's other adjectives i can add here but it just usually is a bunch of ease inactivism is something that natural systems can carry out and it's something that we can grasp at least at the atomic level even simulate sometimes computationally and what is being gained by phrasing the entire system as a control system for action rather than having a so-called detachable interface like we were seeing in the previous approach um by framing explicitly about action you end up training on small or large data sets and evolving iterations of systems with an eye towards function rather than with an eye towards and that can look like a lot of different things in different areas so a few other examples of previous work in the control systems area are like cybernetics also control systems can include elements that are game players they can include uh elements of control that have to do with swarm control as well this was an interesting paragraph from a technical point of view active inference and perceptual inference are not merely two sides of the same coin instead active inference is the name of the formulation for policy selection what advocates of the bayesian brain call perceptual inference is just one moment of the policy selection process in namely state estimation so your your your big deal is just a parameter in our the issue we want to press here repeat is that the active inference framework implies that perception is a form of that is action and perception cannot be pulled apart as they sometimes are in the bayesian brain framework so that's that detachable critique from the inactivist to the um structuralism they say because you're thinking structurally whatever that means you have some detachability between form and function we have a non-distinction between form because there's uh all these different inactive things happening so we're gonna have a better perspective because we can already computationally model our perspective too just like you could so closer to the end of the paper they write in summary the role of a is subtle in active inference the generative model itself never actually exists outside the dynamics that is outside the adaptive actions and policy selection of the organism well and so i'll just note there that adaptive means that there's a striving for success but just because something is adaptive doesn't mean everyone adapts within the dynamics it provides a point of reference or definition of variational free energy or more precisely a definition of the gradients with respect to internal and active states given that the vicarious realization of the generative model through a minimization of the variational free energy can only be through action and changes in internal state we can think of the generative model as being enacted and of the recognition density as being embodied so what is being said here is because there's sort of this thermodynamic forwardness that drives the active inference process as far as far from equilibrium systems doing what they do because those two things are enacted and and because it maps onto some of these previous distinctions that were raised and at least computationally resolved in the computationalist paradigm so we want to draw those mappings about data and about higher order perceptions of data map that onto our understanding of the world but not end up computationalizing the world per se that's one implication there's probably others and then they say this speaks directly to embodied and inactive approaches in cognitive neuroscience and provides a computationally tractable framework for the metaphors mobilized by these paradigms so they're again they're saying like we'll be able to computationally so more attractively do some of the stuff that people thought philosophically or ecologically would be cool generative models encode exploitable structural information about the world our analysis suggests that this is false indeed in this article we sought to underpin the claim that generative models do not encode anything directly they are rather expressed in embodied activity and leverage information encoded in the recognition density which is an approximate posterior belief or best guess assuming our conclusion is correct and why wouldn't you our inactive inference proposal serves to free us from a standard but flawed philosophical assumption about the nature and explanatory basis of cognition and adaptive behavior so that's the closing words of the paper and they come out hard with the implications of action on the bayesian computationalist worldview let's look a little bit about how active steps into that gap between the agent world gap and then also between the bayesian computationalist and then the more inactive and embedded approach so here's another quote from paper they write in active inference under the fep which we'll again we'll return to the generative and recognition models are best cast as realizing inference and control the belief guided selection of action policies so again more like cybernetics less like image classifiers and do not have the properties ascribed by structural representationalists again read them if you want to figure out exactly what you think they actually are talking about we thus provide a philosophical and information theoretic justification for an inactive view of generative models under the fep so if you read the paper you might learn some more details about why they argue that the generative model is an activist in active inference under the auspices of the fep but here's what they end up coming to they end up linking these two states now i've actually not put any labels on either of these two nodes to emphasize that which the there's a wide range of relationships that the nodes can have because it's again a general graphical model and also it's just going to depend on the scientists and how it's being modeled um from a scientific instrumentationalist point of view and then if you're talking about organism and niche or organism and organism like improvisation then again it depends on the situation and how you're going to be measuring it so we'll just leave the labels off but the model that was like the one that goes from the um data like the retinal data to the higher order model model of the world is still in orange on the top and that's the generative model or the and then the recognition models are um actually this one might be flipped i'm not exactly sure but so sorry if the colors are messed up on this one slide but the key aspect is that inference is um what's happening from the data to the higher order priors of the world on the top in orange and so that's like a signal processing which lends itself naturally to predictive processing and all these other things we'll get to and then on the bottom you have on the recognition models side you have control so these models are not just being used to recognize stimuli in terms of making structural representations of the world or updating them but rather in terms of their embodied and inactive control policies so because of this inactive insight or interface in the pragmatic term the fep frames itself especially in this multi-level embedded way as being something that integrates the two models by allowing hopefully the mathematics to be partitioned the way that we expect and would hope that these kind of well behaved bayesian models sometimes can behave but also have all the richness of the inactivist and be able to integrate these multi-scale models so we can think of kind of a few generations of the active inference development and just think what are the commonalities here in the different waves of the development and this is uh as per a recent discussion that was really helpful hopefully i'm representing it roughly accurately with uh maxwell and axel and axe and alec and uh in the early phrasings of active inference there was a local like a one step you can think expectation maximization algorithm so only looking one step ahead in time led to relatively local alignment of the model with the sensory data a second iteration or increase in the nuance of active inference came with this paper on inactive inference and others which introduced temporal depth and niche into the equations as well as things like planning and agent embeddedness and and the beginning surveillance and then in some of the more recent work like of 2019 and 2020 there's work related to agent learning and affect and these sophisticated or counterfactual aspects of action and cognition and counterfactuals about belief um all the stuff that i really would love to learn more about so in the next and uh last part of this stream we're gonna talk just first coat of paint about the connection between a few different topics and a few different philosophy of science topics so the first set of topics the scientific topics are the bayesian brain predictive coding active inference and the free energy principle and then we're gonna see how these frameworks or theories or principles or whatever they are we'll define them how are they all linked specifically as theories frameworks hypotheses and connected by falsification or other mechanisms of comparison and then on the right side is uh this uh meme and so on the top right this is sort of reflected uh to get the punch line first here too on the bottom um is a 2018 alias bulletin interview with carl fristin and our departed dearly colleague martin forger and in this interview there's sort of a philosophy of science progression that i'm going to walk through a little bit of today and lay out hopefully to clarify the differences between some of these different theories or what level of science are they really striking at and uh in this meme on the top is the essentialist and the positivist perspective so the essentialist being the idea that the uh objects or things have meaning in themselves not just as in relation with other objects and positivists being the idea that positive claims about nature are made through measurement and science then in the second panel is the utilitarian so defined based upon what they perceive their value functionally to be and then falsificationist which as we'll return to is about a theory that makes a prediction that would make it incorrect or make it uh untenable or increasingly untenable to hold and then the third panel is the pluralist and the bayesian and so in this world view there's understood to be a balance between novelty and conservation and that a somewhat optimal balance for certain types of ultimately utilitarian or pluralist views a certain value parameters can ultimately be struck but there's never going to be a perfect single value because there's still this question about values to bring to the table so not to take it to world modeling but you know you're thinking about these issues as bayesians pluralists and then in this paper fristen just sort of has an another level that uh almost merits this following chart that i'll show that really lays out again what the fep and the active are and how they're related to specific testable hypotheses because i know that's a common thing that people want to talk about so the green is what martin and i had asked we wrote in other words what is explained by the entropic reduction within the free energy principle that is not explained by any model parameters in the other frameworks so this was in an attempt to understand what a lot of people are trying to understand which is what are we getting above and beyond as a value add or conceptually or utilitarian perspective whatever it is just what are we getting by modeling this quantity as you've laid it out in these certain works and here's what fristen responded i think it is useful to make a fundamental distinction at this point that we can appeal to later the distinction is between a state and a process theory i.e the difference between a normative that things may or may not conform to in a process theory or hypothesis about how that principle is realized under this distinction the free energy principle stands out in stark distinction to things like predictive coding and the bayesian brain hypothesis this is because the free energy principle is what it is a principle like hamilton's principle of stationary action it cannot be falsified so that was obviously uh an almost inflammatory stance for many they uh fainted when they heard first then say that fap cannot be falsified it cannot be disproven in fact there's not much you can do with it uh that's you know the other choice sound bite right there's not much you can do with it oh well then what are we doing here right unless you ask whether measurable systems conform to the on the other hand hypotheses that the brain performs some sort of bayesian or predictive coding are what they are hypotheses these hypotheses may or may not be supported by empirical evidence so i'm going to just lay out this chart on this continuum to get at the distinction between all these different philosophy of science terms principle theory framework everything so this is going to be a continuum on the bottom on the x-axis from the left side more theoretical or abstract moving through the scientific endeavor to the statistical and the applied so this isn't uh a ranking so much as it is a continuum just from abstract to uh practice based and individual people can can collaborate and exist across multiple layers of wanting to understand how this all and as sort of laid out in what i'll present as a slightly clarified and i believe a little bit interpreted uh unpacking of what fristen wrote in that i'm gonna try to lay it out very clearly uh there's some gray area but i think this is just one way to lay it out so towards the most theoretical and abstract side we have principles which are axioms and invariances so these are some of the things that you can't falsify and i thought a lot a lot about that when i was thinking about what to say about this and one thing that it made me think about was girdle escherbach the 1979 book by hofstetter which is a really exciting exploration about a lot of these topics related to axioms and logic and also it reminded me of the law of conservation of stoichiometry in chemistry so not just the law of conservation of mass or mass but of stoichiometry and chemistry if you'll remember from taking a course it's about balancing the equations and so it's a rule within a certain framework of problem solving that equations have to be balanced now there's also a slightly stronger more general belief that there's a conservation unless there's a nuclear reaction or an alchemical transformation that there's a uh chemical elements that because such a strong conservation at the nano level that there's also some sort of generalized conservation at a higher level so that's a belief in a principle and that ends up framing what you see how you see it how you measure it how you compare it it tends to be normative which means using like a should or an ought like oh well this motor ought to waste heats relative to 100 efficient motor so i think we can pretty much detach from the uh some of the emotional psychological baggage on that and just think about the scientific normativity and then what it actually says and then our response to it which is absolutely valid and important is something we can work with and grow from the principle doesn't need to apply to all scales so the laws that say the patterns that say that the number of molecules in a chemical reaction are conserved it doesn't need to say everything about social interactions so they don't need to be at all scales sometimes they're sort of incorrectly called laws i'm just not really into thinking about nature is having laws i think laws are something that people make and then there's patterns in nature and whose last name you attach to which law is just wikipedia url bragging rights what matters is the patterns that we're discovering so again conservation of mass energy pattern law and this kind of returns us to this dual instrumentalism of principles like the free energy principle which is that on one hand it can be about the system like you can choose to model the system as neither gaining or losing energy or as always decreasing free energy or you could make a stronger claim that the principle is actually how the system is in which case you're saying that it is the system's functional principle that it is under national selection or it does follow the second law of thermodynamics all right so drilling down a little bit towards the scientific and the statistical a theory is a coherent set of ideas claims in logic and it makes specific testable hypotheses that are drawn from theory that's upstream and so that's sort of the confusing thing is from uh abstractness uh ranking the principle is the the ground uh the bedrock of the theory the water that theory swims in but it's more on the theoretical side so theory isn't even the most theoretical thing and then people say oh yes i do theory or something like that and that can mean a broad range of things from anywhere from the middle to the uh theory when it's partnered with tools and approaches is a framework also community of practice is important but frameworks are important too and the way that theories are compared is often through falsification or theory comparison so falsification is just like the best model in play is the one that we go with if there's some metaphor out there about hunting ducks or something with a shotgun i don't know and uh there's other mechanisms of theory comparison theory anarchy all these wild thoughts but um the falsificationist worldview that like we go with what works and then end up having a paradigm shift and moving into a new set of dorms around a scientific topic like landslide either in a little micro sense or in a larger macro sense all this kind of stuff there's a lot of depth i i'm just going to briefly just say people have thought about these kind of theory comparators now in the alias interview response firstin mentioned two kinds of theories he mentioned state theory which he used to mean something pretty similar to a normative principle those were about system states and then he combined the normativity with it being about the system states so it said that there was a normative stance that the system will conform to a certain way of being and it's about the states and that was contrasted with the process theory um which is about how systems change which is why it's called a process theory all right now to get to a specific hypothesis a hypothesis is a specific prediction about the system that's under the auspices of a specific uh theory whether it's a state theory or a and a hypothesis is tested with approaches it can also be explored with visualization so the hypothesis is tested from a falsificationist paradigm at the quantitative level so at the confidence level of 0.01 or 0.001 or according to this metric those are quantitative sort of in the trenches falsificationism of null hypothesis which is that there's no influence of a certain treatment for example whereas the theory falsificationism that's like one layer higher up is a little bit more um like logical uh battles being fought no single t-test brings down a theory but can take down a hypothesis and so let's break this down for the free energy principle specifically so as fristen had mentioned the free energy is this principle and i think in future times it would be cool to talk about what makes it a principle what are the assumptions what are the implications what are related assumptions and for now we'll just say that downstream of the free energy principle is active inference active inference is um a process theory it's a coherent set of ideas claims and logic about the world it is going to be making specifically testable hypotheses specifically about the structure of action and about how agents interact with their environment and examples of theories that have overlap with active inference um for example licensing the use of the people who are structural representationalists using the bayesian brain metaphor to think about the bayesian brain as a theory we can think about the hypotheses that are specifically made by that theory so for example the claims might be something like at the first level that the brain can be modeled with bayesian stats and then at the second level like that the brain implements bayesian statistical algorithms in what it does in nature so that's that dual instrumentalism and then another area of theories that has overlap with active inference as well as the bayesian brain just didn't want to clutter it was predictive coding and two hypotheses in predictive coding are one would be sense data can be analyzed relative to now casted predictions now casting and the second level would be something like organismal sensory relays are predictive so again it's the dual instrumentalism where there's one level about how the scientist analyzes the world which is that yes we can use this framework to understand data even data about organisms or data about simulations or systems and then at the second level it's actually the claim about the world and so in both of these cases their hypotheses because they are under an axiomatic framework reflected by the free energy principle that's why it keeps on saying inactive inference under the free energy and the active inference framework also draws on not just the bayesian brain predictive coding but just to label one more up there is there's this inactive area that we talked a lot about today and all the cool insights that the inactive side has to bring to the table so maybe another time we could think about what the principles underlying and activism are or are there fundamentals it's an air i'd love to learn more about are there frameworks for inactive inference i know that we've had some participants on the active stream who had a lot more experience and practice with these kinds of things so we want to see how we can make active inference this hub for realizing the ideas and the people and the projects coming together to kind of make it real to show how these ideas are all connected and uh yeah see where we can kind of take these ideas just conceptually and from an impact perspective so that's the end of 6.0 thanks for participating we provide follow-up forms to live participants it'll be pretty easy this one any feedback suggestions or questions would be totally welcome stay in communication with us and yeah the next uh live stream will be 6.1 that is going to be on october 13th on the tuesday then we are going to have 6.2 on the 20th that's going to be a follow-up discussion so if you're interested in 7 a.m pacific time for 60 to 90 minutes on october 13th or october 20th those are going to be discussing this paper a tale of two densities then on october 27th and november 3rd both tuesdays same time we're going discussing another paper you can check it out on our twitter if you want to learn more and if you're watching this after those dates thanks for watching please feel free to still leave a comment and uh yeah thanks a lot watching just let us know if you have any thoughts or questions but have a good weekend or whenever it is you're listening bye